

Microplastic Degradation Research Project

Phase 3 — Engineered IsPETase Variant Design

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This document records Phase 3: applying eight targeted mutations to the wild-type IsPETase sequence with full pre and post verification, producing the engineered gastric-stable variant.

The 8 Mutations — Full Table

Pos	WT	Mut	Category	Verified	Rationale
51	T	C	Disulfide	OK	Forms S-S bond with T72C, Cb-Cb = 4.50 Å
72	T	C	Disulfide	OK	Partner of T51C disulfide bond
59	R	E	Charge reversal	OK	Highest surface exposure <i>rel_ASA</i> =0.68, 26.0 Å from triad
117	L	I	Pepsin	OK	Only surface-exposed high-risk pepsin site
132	R	E	Charge reversal	OK	Surface Arg, <i>rel_ASA</i> =0.40, 22.3 Å from triad
224	R	E	Charge reversal	OK	Surface Arg, <i>rel_ASA</i> =0.54, 22.7 Å from triad
280	R	E	Charge reversal	OK	Surface Arg in subsite IIc, <i>rel_ASA</i> =0.48, 15.0 Å from triad
285	R	E	Charge reversal	OK	Surface Arg, <i>rel_ASA</i> =0.40

All Verification Checks

Check	Result	Status
WT residues verified	All 8 match	OK
Mutations applied	8 of 8	OK
Sequence length	290 aa	OK
Catalytic Ser160	S — unchanged	OK
Catalytic Asp206	D — unchanged	OK
Catalytic His237	H — unchanged	OK
Net charge WT at pH 1.2	+22e	OK
Net charge ENG at pH 1.2	+17e	OK
Total charge shift	-10 units	OK

All 9 verification checks passed. Catalytic triad is fully intact. Net surface charge reduced by 10 units at pH 1.2 through five Arg to Glu charge-reversal substitutions. Engineered FASTA saved and ready for Phase 4.